

Plots

August 24, 2026

```
[2]: import numpy as np
      %matplotlib widget
      import matplotlib.pyplot as plt
      import matplotlib.cm as cm
      import matplotlib.colors as mcolors
      from scipy.optimize import curve_fit

      from pathlib import Path

      base = Path.cwd()
      Ausgabe_path = Path(base/"Diagramme")
      print(Ausgabe_path)
```

c:\Users\samue\OneDrive\Dokumente\FP01\Diagramme

```
[3]: def fit_func(f,V,W1,W2,n1,n2):
      f_safe = np.clip(np.abs(f), 1e-12, None)
      return V/(np.sqrt(1+1/(f/W1)**(2*n1))*np.sqrt(1+(f/W2)**(2*n2)))
      def fit_func2(f,V,W1,n1):
      f_safe = np.clip(np.abs(f), 1e-12, None)
      return V / np.sqrt(1 + 1 / (f_safe / W1) ** (2 * n1))
```

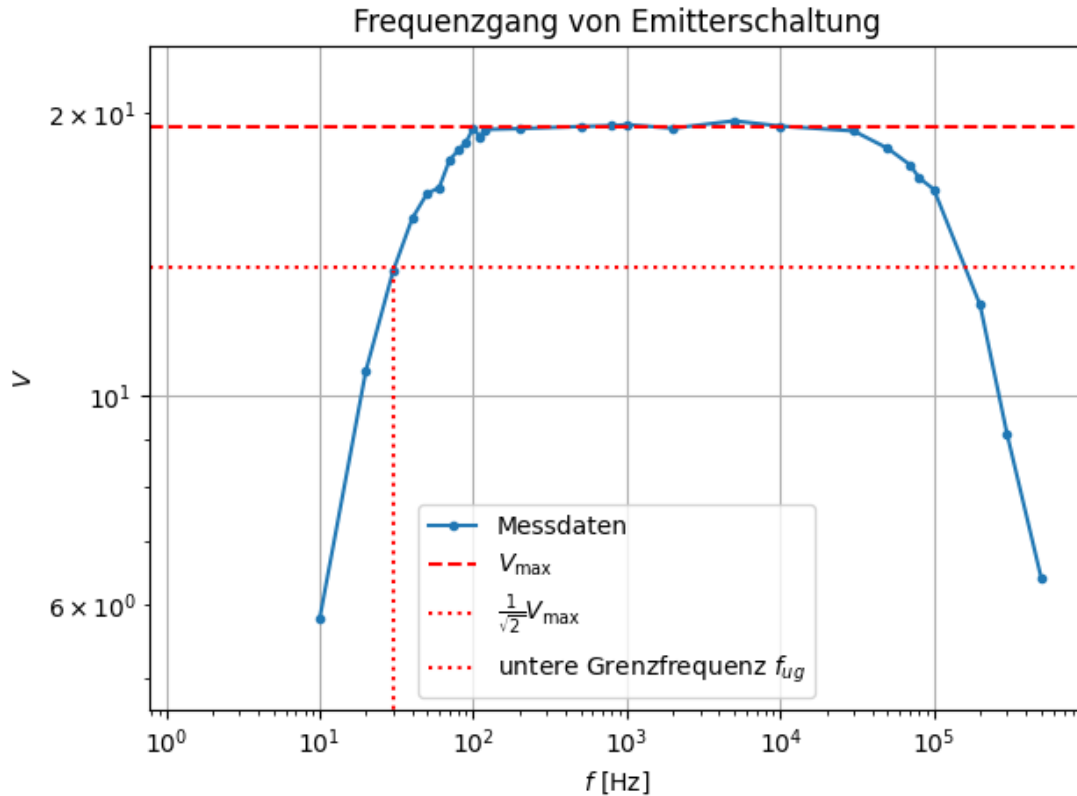
```
[4]: plt.close('all')
      plt.figure()
      plt.title(r'Frequenzgang von Emitterschaltung')
      plt.xlabel(r'$f$ [Hz]')
      plt.ylabel(r'$V$')
      f=np.concatenate((np.
        ↪array([10,20,30,40,50,60,70,80,90,100,110,120,200,500,800]),np.
        ↪array([1,2,5,10,30,50,70,80,100,200,300,500])*1e3))
      V=np.array([5.79,10.64,13.57,15.45,16.43,16.67,17.83,18.28,18.64,19.31,18.90,19.
        ↪23,19.27,19.37,19.41,19.46,19.29,19.64,19.39,19.16,18.36,17.63,17.08,16.
        ↪60,12.5,9.1,6.4])
      # D=1e-3
      # g=P/D
      plt.loglog(f,V,marker='.',label=r'Messdaten')
      plt.axhline(np.mean(V[11:
        ↪19])),color='red',linestyle='dashed',label=r'$V_{\text{max}}$')
```

```

plt.axhline(1/np.sqrt(2)*np.mean(V[11:
    ↳19]),color='red',linestyle='dotted',label=r'\frac{1}{\sqrt{2}}V_{\text{max}}$')
plt.
    ↳vlines(x=f[2],ymin=0,ymax=V[2],color='red',linestyle='dotted',label=r'untere_
    ↳Grenzfrequenz $f_{\text{ug}}$')
# plt.vlines(x=37.
    ↳4,ymin=0,ymax=V[3],color='red',linestyle='dotted',label=r'$theoretische_
    ↳Grenzfrequenz$')

plt.ylim(np.min(V)*0.8,np.max(V)*1.2)
# def fit_func(f,V,W1,W2,n1,n2):
    # return V/(np.sqrt(1+1/(f/W1)**(2*n1))*np.sqrt(1+(f/W2)**(2*n2)))
# p0 = [1000 , 1000 , 50000 , 5 , 5]
# popt, pcov = curve_fit(fit_func, f, g ,p0)
# plt.loglog(f, fit_func(f, *popt), label='Fit')
# for i in range(len(p0)):
    # wert,fehler,latex,_=round_to_sigs(popt[i],np.
    ↳sqrt(pcov[i][i]),einheiten='')
    # print(latex)
    # print(fehler/wert*100)
plt.legend()
plt.grid()
plt.tight_layout()
plt.show()

```



```
[5]: popt, pcov = curve_fit(fit_func, f, V ,p0=[20, 10000, 100000, 1, 1])
```

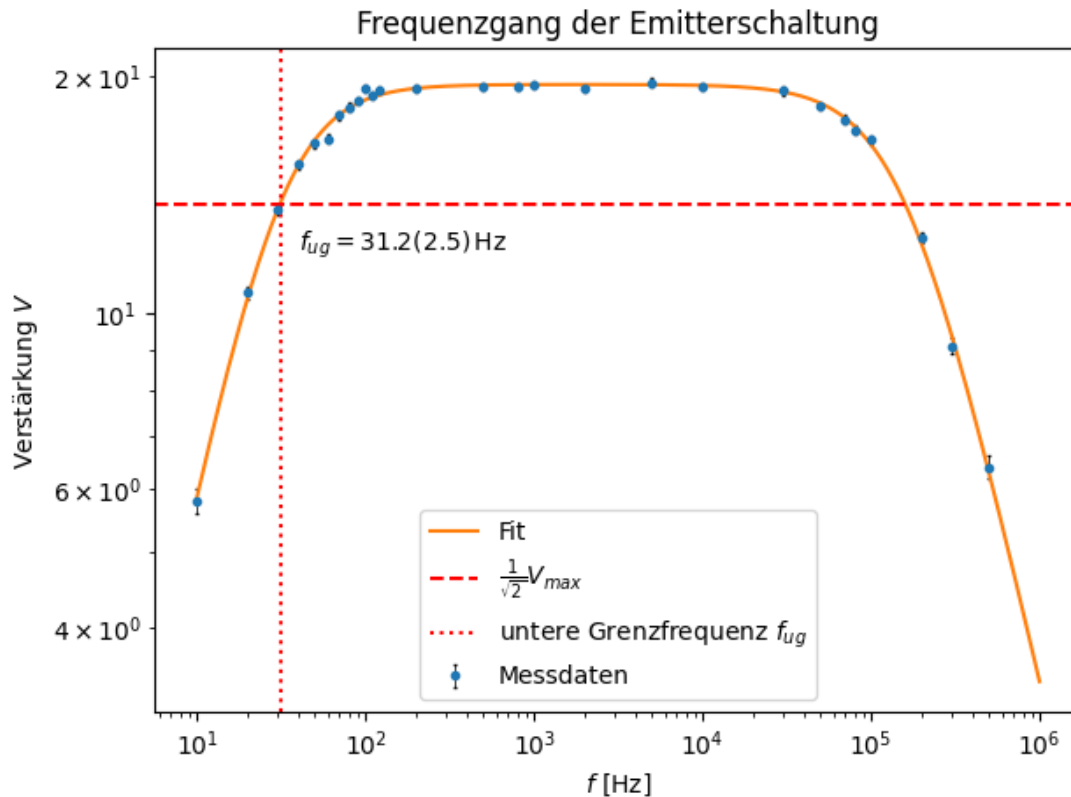
C:\Users\samue\AppData\Local\Temp\ipykernel_7396\2784924508.py:3:

RuntimeWarning: invalid value encountered in power

```
return V/(np.sqrt(1+1/(f/W1)**(2*n1))*np.sqrt(1+(f/W2)**(2*n2)))
```

```
[30]: x=np.logspace(1, 6, 1000)
plt.figure()
plt.title(r'Frequenzgang der Emitterschaltung')
plt.errorbar(f,V,yerr=0.21,fmt='.',elinewidth=0.
    ↳5,capsize=1,ecolor='black',label=rf'Messdaten')
plt.plot(x, fit_func(x, *popt), label='Fit')
plt.axhline(np.max(fit_func(x, *popt))/np.sqrt(2),
    ↳label=r'$\frac{1}{\sqrt{2}}V_{\max}$', color='red',linestyle='dashed')
plt.axvline(x=31.2, color='red', linestyle='dotted',label='untere Grenzfrequenz
    ↳$f_{\text{Ug}}$')
plt.text(4e1, 12, r'$f_{\text{Ug}}=31.2(2.5) \text{ Hz}$', color='black')
plt.xlabel(r'$f$ [Hz]')
plt.ylabel(r'Verstärkung $V$')
plt.xscale('log')
plt.yscale('log')
```

```
plt.legend()
plt.tight_layout()
plt.show()
plt.savefig(Ausgabe_path/"Frequenzgang_Fit_Emitterschaltung.
↳png",format="png",bbox_inches='tight')
```



```
[7]: print(popt)
```

```
[1.95208965e+01 3.12147767e+01 1.58777660e+05 1.01705688e+00
 9.36958459e-01]
```

Frequenzgang Kollektorschaltung

```
[8]: f_col=np.
↳array([10,30,50,70,100,500,2000,5000,10000,30000,50000,70000,100000,300000,500000])
↳#Hz
V_col=np.array([0.2, 0.7, 0.9, 0.9, 0.9, 0.96, 0.94, 0.99, 0.95, 1.02, 0.99, 1.
↳13, 0.95, 0.95, 0.96]) #Verstärkung
V_col_f=np.concatenate((np.full(5, 0.5), np.full(10, 0.10))) #Fehler
```

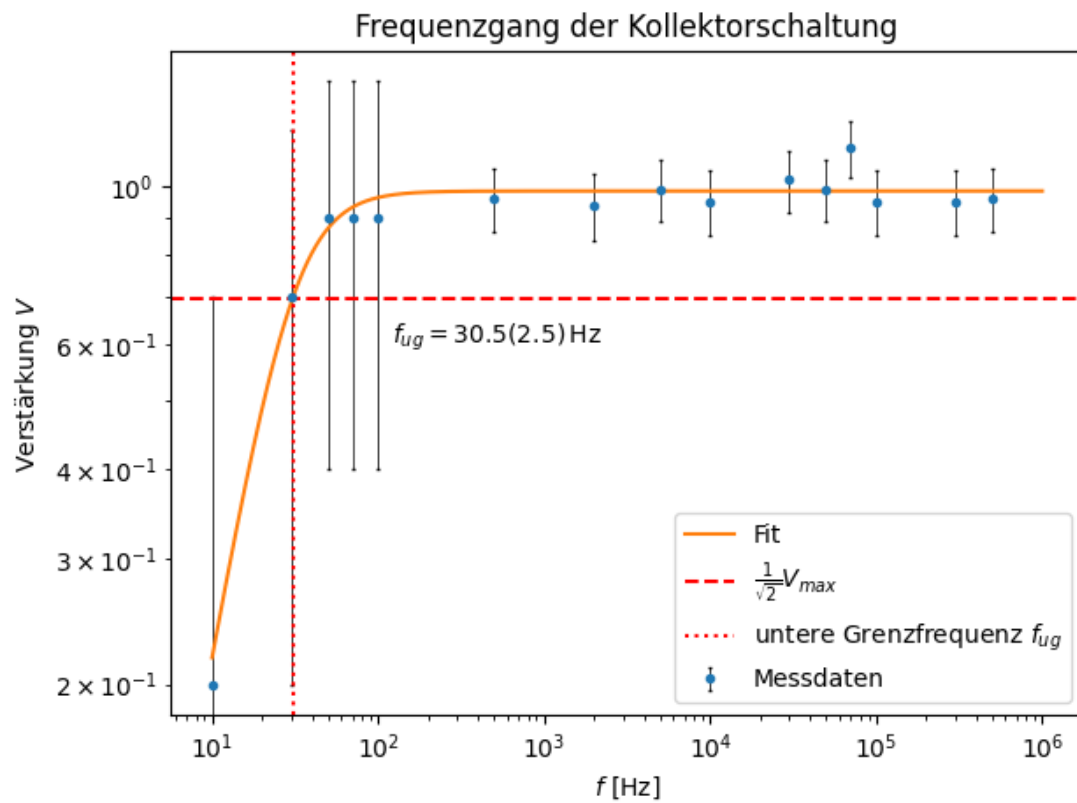
```
[9]: popt1, pcov1= curve_fit(fit_func2, f_col, V_col, sigma=V_col_f,
    ↪absolute_sigma=True, p0=[1, 10000, 1])
x2=np.logspace(1, 6, 1000)
```

C:\Users\samue\AppData\Local\Temp\ipykernel_7396\2784924508.py:6:

RuntimeWarning: invalid value encountered in power

```
return V / np.sqrt(1 + 1 / (f_safe / W1) ** (2 * n1))
```

```
[31]: plt.figure()
plt.title(r'Frequenzgang der Kollektorschaltung')
plt.errorbar(f_col, V_col, yerr=V_col_f, fmt='.',elinewidth=0.
    ↪5,capsize=1,ecolor='black', label=rf'Messdaten')
plt.plot(x2, fit_func2(x2, *popt1), label='Fit')
plt.axhline(np.max(fit_func2(x2, *popt1))/np.sqrt(2),
    ↪label=r'$\frac{1}{\sqrt{2}}V_{\max}$', color='red',linestyle='dashed')
plt.axvline(30.5, color='red', linestyle='dotted',label='untere Grenzfrequenz
    ↪$f_{\text{ug}}$')
plt.xlabel(r'$f$ [Hz]')
plt.ylabel(r'Verstärkung $V$')
plt.text(1.2e2, 6e-1, r'$f_{\text{ug}}=30.5(2.5) \text{ Hz}$')
plt.xscale('log')
plt.yscale('log')
plt.tight_layout()
plt.legend()
plt.savefig(Ausgabe_path/"Frequenzgang_Fit_Kollektorschaltung.
    ↪png",format="png",bbox_inches='tight')
```



[]: